Afternoon Lab - Commissioning & Fault Finding

PV101 Day 5 - Monday, July 13, 2026

Lab purpose	**Use the completed Day 4 systems as commissioning and troubleshooting trainers. Students inspect, measure, diagnose, document, and restore.**
Team setup	Five systems for six students. One two-person team is appropriate. Rotate roles so each student meters, documents, and explains evidence.

Roles

Role	Job	Required statement
Lead inspector	Traces path, checks labels, fuse placement, conductor condition.	I can point to every source and protection point.
Meter lead	Selects range/ports and takes required readings.	I know the expected reading before I measure.
Recorder / verifier	Records evidence, watches sequence, confirms restoration.	We change one thing and verify the result.

Part A - Commission the existing build

- Do not dismantle the system. Begin with a visual inspection.
- Draw or trace: PV -> controller -> battery -> fuse/protection -> inverter/load.
- Circle every current source and star every protection point.
- Check polarity labels, exposed copper, strain relief, fuse access, and serviceability.
- Record battery open-circuit voltage: _____ V.
- Record controller battery-terminal voltage: _____ V.
- Record PV/input voltage when safe and available: _____ V.
- Record load/inverter input voltage under load: _____ V.
- Record load-end voltage under load: _____ V.
- Voltage drop = source-end voltage - load-end voltage: _____ V.
- Instructor initials for commissioned system: _____.

Part B - Fault diagnosis

Fault card or station: _____

Observed symptom: _____

Expected normal condition: _____

Prediction before measuring: _____

First measurement and result: _____

Second measurement and result: _____

Fault identified: _____

Single corrective action: _____

Evidence system returned to normal: _____

Part C - Wire and fuse reasoning

- Assigned or measured load power: _____ W.
- System voltage under load: _____ V.
- Estimated current, A = W / V: _____ A.
- Installed or proposed conductor size: _____.
- Installed or proposed fuse size: _____.
- Explain why the fuse protects the conductor: _____
- Would a longer wire run improve or worsen voltage drop? Explain: _____

Part D - Serviceability handoff

- Add or correct one useful label.
- Identify one future troubleshooting test point.
- Confirm fuse access and battery isolation method.
- Leave a brief note for the next technician: _____.

Checkout rubric

Skill	Score
Traces complete power path	0 1 2
Predicts expected reading before measuring	0 1 2
Uses meter safely and correctly	0 1 2
Diagnoses from evidence rather than guessing	0 1 2
Explains fuse and conductor relationship	0 1 2
Measures and explains voltage drop	0 1 2
Restores and verifies known-good condition	0 1 2
Improves labeling/serviceability	0 1 2

Exit ticket

One reading that proved something today: _____

One fault I can now diagnose systematically: _____

One reason not to rebuild a working system just to practice: _____